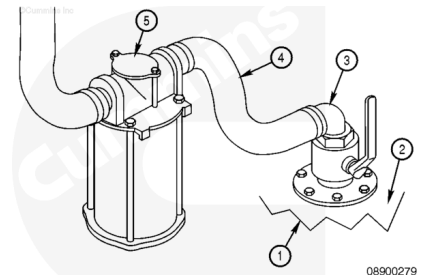


Trainings ▾

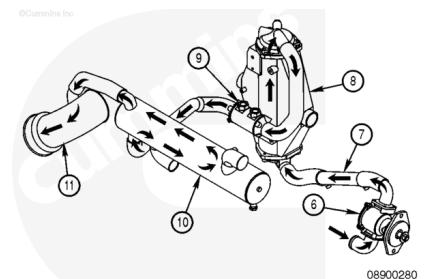
General Information

Typical sea water cooling system

1. Grate-type strainer located on the bottom of the boat hull
2. Vessel bottom/hull
3. Sea water inlet valve
4. Hose
5. Sea water strainer

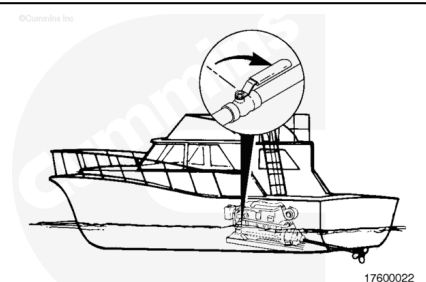


6. Sea water pump
7. Fuel cooler
8. Aftercooler
9. Transmission oil cooler
10. Heat exchanger
11. Sea water outlet.



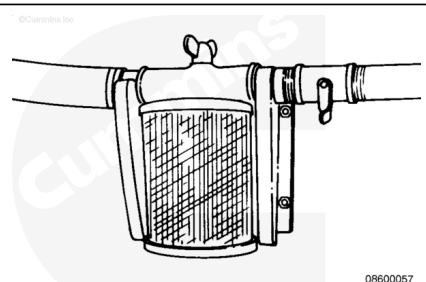
Initial Check

Locate and verify that the sea water inlet valve is in the full open position. If it is closed or partially closed, open the valve and recheck the engine temperature.



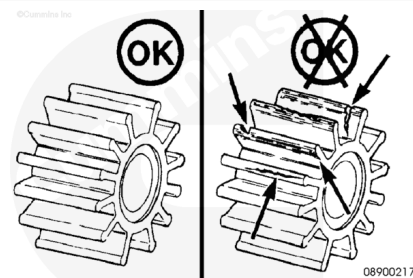
Inspect the sea water strainer for foreign objects that could restrict the water flow.

Some strainers have clear covers for easy inspection. If the strainer has to be opened for inspection, refer to Procedure 008-067 (</qs3/pubsys2/xml/en/procedures/99/99-008-067.html>).

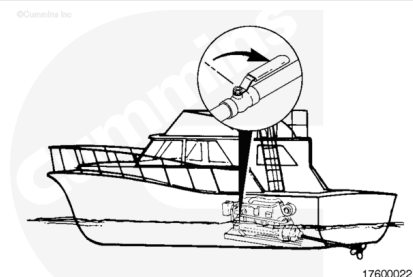


If the engine was run with the sea water system highly restricted due to a closed off sea water inlet valve or a clogged sea water strainer, it is advisable that the sea water impeller **must** be inspected for damage.

If the age of the sea water impeller is unknown, inspection of the sea water impeller is advisable. Refer to Procedure 008-057 (</qs3/pubsys2/xml/en/procedures/41/41-008-057.html>).

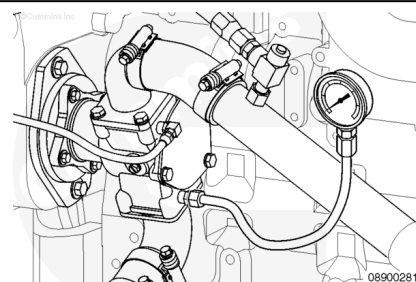


If the engine continues to overheat, perform the following procedures. If the engine does **not** overheat at the dock, perform a sea trial and check for overheating under way.

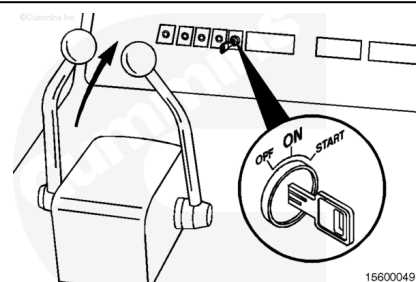


Test

Attach a vacuum gauge to the inlet side of the sea water pump.

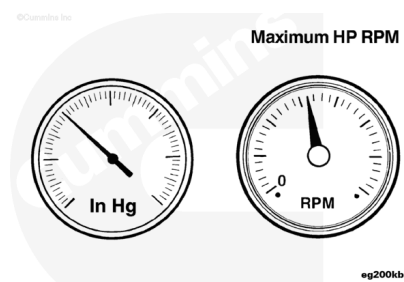


Start the engine.

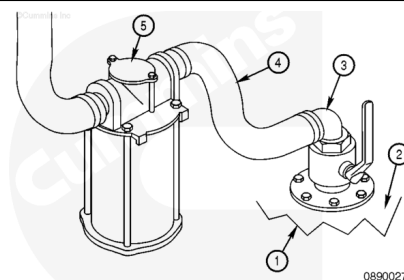


Record the sea water inlet restriction from low idle to rated speed at 500 rpm increments. This test can be conducted while the vessel is at the dock and **not** in gear or underway.

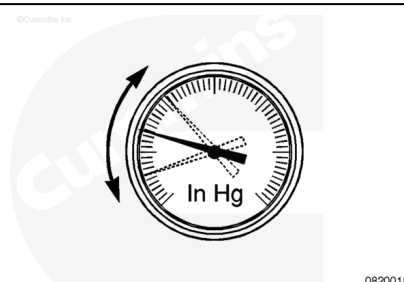
Refer to Procedure 018-018
(/qs3/pubsys2/xml/en/procedures/41/41-018-018-tr.html) for
Marine engine specifications.



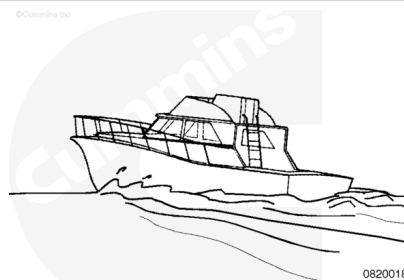
If the sea water inlet restriction is above specification, the source of the restriction **must** be found. Troubleshooting and repair of excessive sea water restriction is a boat manufacturer or boat owner responsibility. Refer to the vessel's OEM service manual. If the inlet restriction is within the specification and the complaint can **not** be verified at the dock, the test **must** be repeated underway.



If the sea water inlet restriction readings fluctuate during the test, inspect the sea water supply for leaks or air intrusion. Troubleshooting and repair for sea water aeration is **not** covered under Cummins warranty.

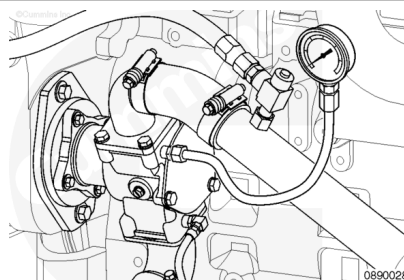


It is possible that aeration (gauge fluctuation) will **only** occur while the vessel is underway due to the introduction of air. Refer to the vessel's OEM service manual.



Pressure Test

Attach a pressure gauge to the outlet side of the sea water pump to check pump outlet pressure.

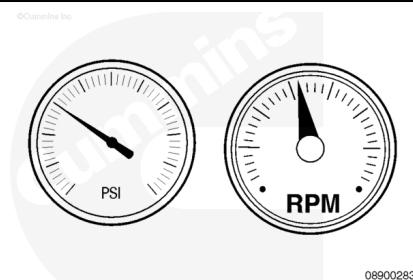


Start the engine and record the sea water outlet pressure from low idle to rated speed at 500 rpm increments. This test can be conducted while the vessel is at dock and **not** in gear or underway.

Refer to Procedure 018-018

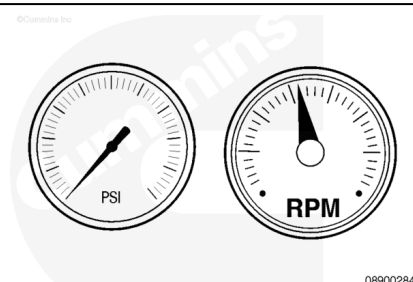
(/qs3/pubsys2/xml/en/procedures/41/41-018-018-tr.html) for Marine engine specifications.

If the sea water pump outlet pressure is within specifications, refer to the Temperature Differential Test in this procedure.



If sea water pressure is **not** present, check the pump for damage. Refer to Procedure 008-057

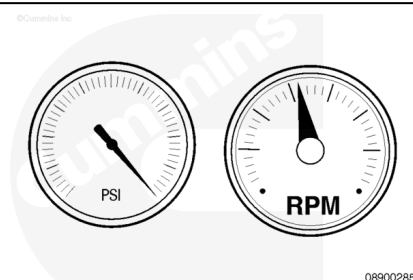
(/qs3/pubsys2/xml/en/procedures/41/41-008-057.html).



If the sea water pump outlet pressure is above the maximum specification test the individual sea water system components for excessive pressure drop as described in the Pressure Differential Test step.

Refer to Procedure 018-018

(/qs3/pubsys2/xml/en/procedures/41/41-018-018-tr.html) for Marine engine specifications.

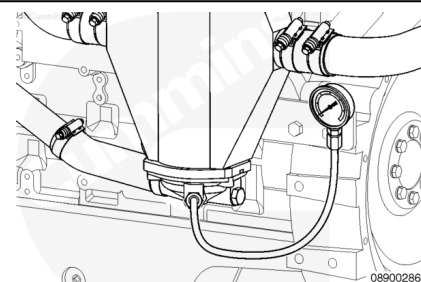


Pressure Differential Test

Check the sea water pressure at the inlet side of the aftercooler at the rated rpm. Record the reading. If the pressure drop between the sea water pump outlet and the aftercooler inlet exceeds the maximum specification, check or replace fuel cooler plumbing as necessary.

Refer to Procedure 018-018

(/qs3/pubsys2/xml/en/procedures/41/41-018-018-tr.html) for Marine engine specifications.

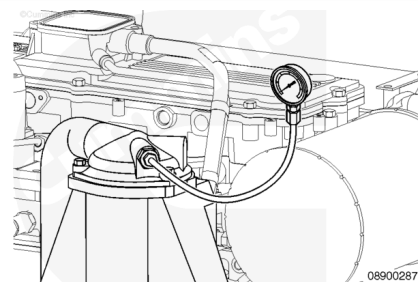


If the pressure drop is within specification, check the sea water pressure at the discharge side of the aftercooler outlet. Record the reading.

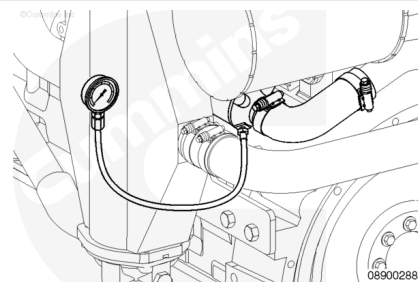
If the pressure drop between the inlet side of the aftercooler exceeds the maximum, check for blockage in the aftercooler. Clean or replace if necessary.

Refer to Procedure 018-018

(/qs3/pubsys2/xml/en/procedures/41/41-018-018-tr.html) for Marine engine specifications.



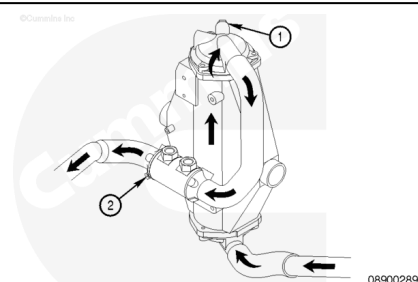
If the pressure drop between the inlet side of the aftercooler and the outlet side of the aftercooler is within specification, attach the pressure gauge to the gear oil cooler drain plug located on the outlet side and check for pressure drop across the gear oil cooler.



If the pressure drop between the inlet side (1) and the outlet side (2) of the gear cooler is greater than the maximum specification, check for blockage in the gear cooler. Clean or replace if necessary.

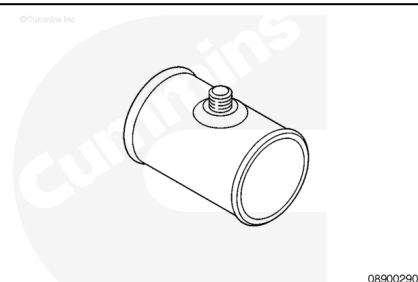
Refer to Procedure 018-018

(/qs3/pubsys2/xml/en/procedures/41/41-018-018-tr.html) for Marine engine specifications.



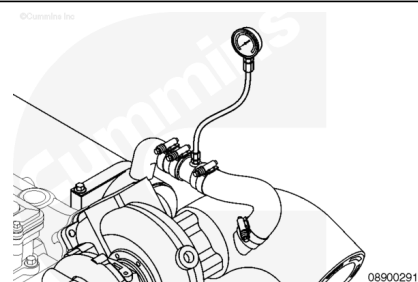
If the pressure drop in the gear oil cooler is within specification, test the pressure drop across the engine heat exchanger.

Fabricate a sea water test tool by using a 44 mm [1 3/4 in] pipe with a fitting in the center to connect a pressure gauge.



Install the fabricated test tool between the outlet side of the heat exchanger and the exhaust elbow. Check the pressure drop across the heat exchanger.

If the pressure drop is greater than the maximum specification, check for blockage in the heat exchanger. Clean or replace if necessary.



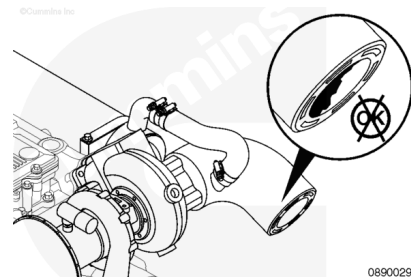
Refer to Procedure 018-018

(/qs3/pubsys2/xml/en/procedures/41/41-018-018-tr.html) for Marine engine specifications.

If the pressure drop across the heat exchanger is within specification, determine the pressure drop across the exhaust elbow (diffuser). This is done by subtracting the heat exchanger outlet pressure from the sea water pump outlet pressure. If this pressure exceeds the maximum specification, check for blockage in the exhaust elbow (diffuser) and exhaust system of the vessel. Clean or replace if necessary.

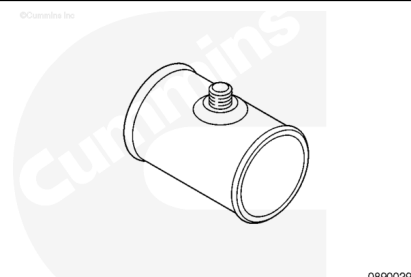
Refer to Procedure 018-018

(/qs3/pubsys2/xml/en/procedures/41/41-018-018-tr.html) for Marine engine specifications.

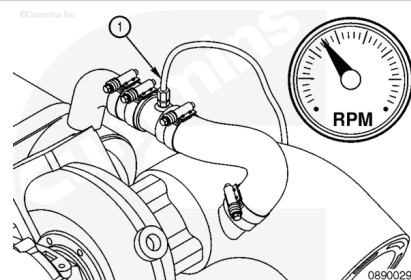


Temperature Differential Test

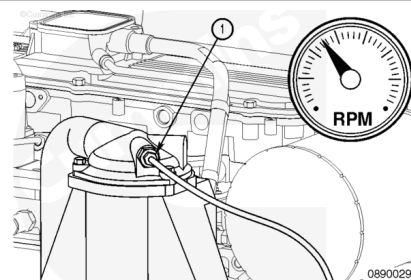
Fabricate a sea water test tool by using a [1 3/4 in] pipe with a fitting in the center to connect a temperature probe.



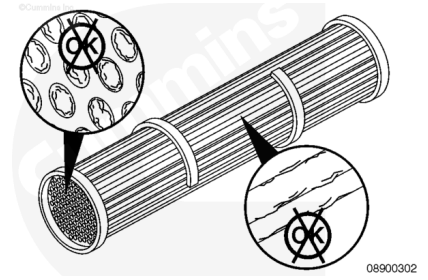
Install the sea water test tool with a temperature probe between the outlet side of the heat exchanger and the exhaust elbow (1). Operate the engine at rated rpm and load and record the temperature.



Install the temperature probe in the outlet side of the aftercooler (1). Operate the engine at the rated rpm and load and record the temperature. If the temperature difference between the aftercooler outlet and the engine heat exchanger outlet is greater than 20° C [40° F], check the sea water pump for water flow problems. Refer to Procedure 008-057 (/qs3/pubsys2/xml/en/procedures/41/41-008-057.html).



If the sea water temperature difference is less than 3°C [5°F], check the heat exchanger for possible plating to the heat exchanger core. Excessive plating or coating to the inside or outside of the cooling tubes will affect the efficiency of the heat exchanger.



Last Modified: 08-Dec-2004
